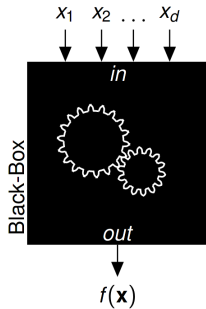
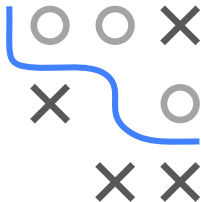


# Optimization in Machine Learning

## Bayesian Optimization

### Black Box Optimization



#### Learning goals

- Definition and properties
- Examples
- Naive approaches

# STANDARD VS. BLACK-BOX OPTIMIZATION

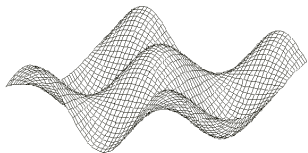
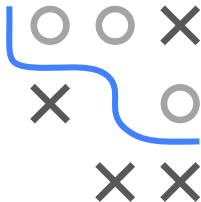
**Optimization:** Find

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x})$$

with objective function

$$f : \mathcal{S} \rightarrow \mathbb{R},$$

where  $\mathcal{S}$  is usually box constrained.



If we are lucky ...

- ... we have an analytic description of  $f : \mathcal{S} \rightarrow \mathbb{R}$
- ... we can calculate gradients and use gradient-based methods (e.g. gradient descent) for optimization

# STANDARD VS. BLACK-BOX OPTIMIZATION

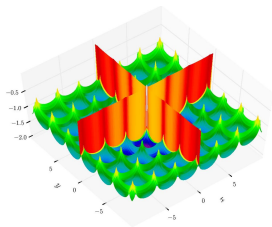
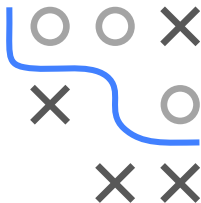
**Optimization:** Find

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x})$$

with objective function

$$f : \mathcal{S} \rightarrow \mathbb{R},$$

where  $\mathcal{S}$  is usually box constrained.



Optimization gets harder ...

- ... if we cannot calculate gradients (because  $f$  is not differentiable or  $f$  is not known to us)
- ... but as long as evaluations of  $f$  are cheap, we can use standard derivative-free optimization methods (e.g. Nelder-Mead, simulated annealing, EAs)

# STANDARD VS. BLACK-BOX OPTIMIZATION

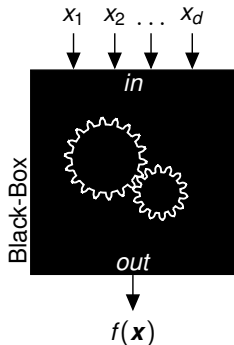
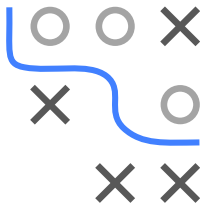
**Optimization:** Find

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x})$$

with objective function

$$f : \mathcal{S} \rightarrow \mathbb{R},$$

where  $\mathcal{S}$  is usually box constrained.



Optimization gets **really hard** if ...

- ... there is no analytic description of  $f : \mathcal{S} \rightarrow \mathbb{R}$  (**black box**)
- ... evaluations of  $f$  for given values of  $\mathbf{x}$  are **time consuming**



# EXAMPLES FOR BAYESIAN OPTIMIZATION

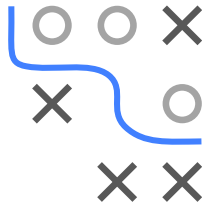
## 2 Optimization of a cookie recipe [▶ Kochanski et al. 2017](#)



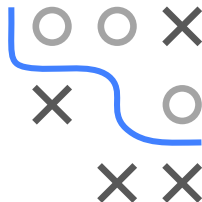
[▶ Click for source](#)

| Ingredient | Salt<br>(tsp)† | Total<br>Sugar (g) | Brown<br>Sugar (%) | Vanilla<br>(tsp)† | Chip<br>Quantity (g) | Chip<br>Type           |
|------------|----------------|--------------------|--------------------|-------------------|----------------------|------------------------|
| Min        | 0              | 150                | 0                  | 0.25              | 114                  | {Dark, Milk,<br>White} |
| Max        | 0.5            | 500                | 1                  | 1                 | 228                  |                        |

- **Goal:** Find “optimal” composition and amounts of ingredients
- **Evaluation:** Cookies are baked according to the recipe, tested and rated by volunteers



# NAIVE APPROACHES

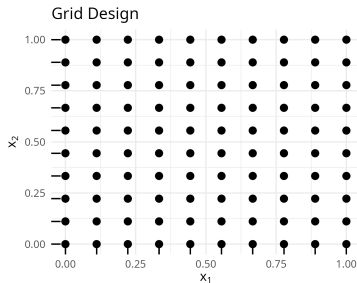
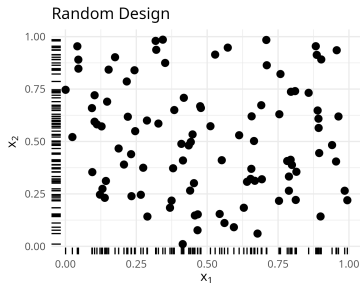
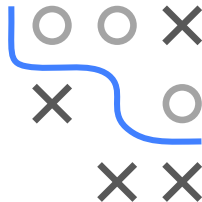


- 1 Empirical knowledge / manual tuning
  - Select parameters based on “expert” knowledge
  - **Advantages:** Can lead to fairly good outcomes for known problems
  - **Disadvantages:** Very (!) inefficient, poor reproducibility, chosen solution can also be far away from a global optimum

# NAIVE APPROACHES

## 2 Random search / Grid search

- Random search: Evaluate uniformly sampled inputs
- Grid search: Exhaustive search of a predefined grid of inputs
- **Advantages:** Easy, intuitive, parallelization is trivial
- **Disadvantages:** Inefficient, search large irrelevant areas

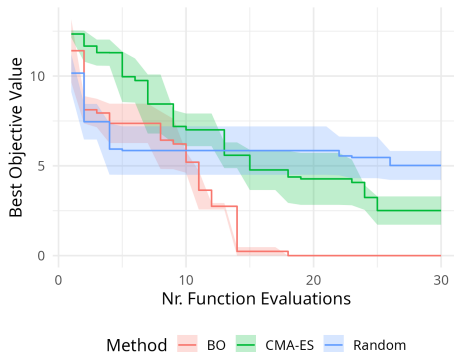
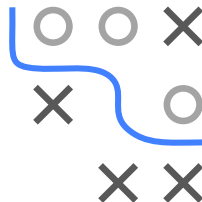


Rug plots of RS vs. GS.

# NAIVE APPROACHES

## ③ Traditional black-box optimization

- Traditional approaches that do not require derivatives
- E.g. Nelder-Mead, simulated annealing, EAs
- **Advantages:** Truly iterative, focuses on relevant regions
- **Disadvantages:** Still inefficient; usually lots of evaluations are needed to produce good outcomes



BO vs. CMAES vs. RS on 2D Ackley.